

# Cybersecurity, IT Act and Security Controls — Detailed Lecture Notes

JKSSB Computer Awareness

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## 56.1 Why security, and the CIA triad

A working computer holds records worth protecting: student marks, account details, personal files, and official correspondence on the office desktop. **Cybersecurity** is the practice of protecting computers, networks, and data against theft, damage, unauthorised access, and misuse. Three goals organise the whole topic, remembered as the **CIA triad: confidentiality** (only authorised persons can read the data), **integrity** (the data is not altered by unauthorised means), and **availability** (the system stays usable when needed). Every threat in this lesson attacks one of these goals, and every control defends one.

**Definition (Cybersecurity).** *The protection of computer systems, networks, and data against theft, damage, disruption, and unauthorised access, so that confidentiality, integrity, and availability are preserved.*

| Term         | Meaning in this lesson                                                          |
|--------------|---------------------------------------------------------------------------------|
| Malware      | Umbrella term for malicious software written to harm or exploit.                |
| Virus        | Malware that attaches to a host file and needs human action to spread.          |
| Worm         | Standalone self-copying malware that spreads over a network unaided.            |
| Trojan horse | Malware disguised as useful software; opens a backdoor; does not replicate.     |
| Ransomware   | Malware that locks or encrypts files and demands payment.                       |
| Encryption   | Coding data so only authorised persons can read it.                             |
| SSL / TLS    | Secure Sockets Layer / Transport Layer Security; secure transmission protocols. |
| Firewall     | Barrier filtering network traffic against rules.                                |
| Quarantine   | Isolation of a suspicious file so it cannot run.                                |
| MFA          | Multi-Factor Authentication; two or more identity checks at login.              |

The rest of this lesson uses these terms in one consistent sense. **Malware** is always the general term, and virus, worm, Trojan, and ransomware are its specific types. **Encryption** always means hiding content (confidentiality), while **digital signature** always means proving the sender and intactness (authenticity and integrity). The exam tests exactly these pairings, so the wording here is kept fixed.

## 56.2 The Information Technology Act, 2000

The **Information Technology Act, 2000** is India's law for the electronic world. Its scope has three parts: it gives **legal validity to electronic records and electronic signatures**, so that online transactions and electronic governance stand on a legal footing; it enables **electronic governance** (official records, notices, and transactions in electronic form); and it defines **cyber contraventions and offences**, such as unauthorised access to a computer, damage to computer systems and data, and fraud or cheating committed through a computer.

**Definition (Information Technology Act, 2000).** *India's legislation of the year 2000 that recognises electronic records and electronic signatures in law, supports electronic governance and electronic commerce, and provides for cyber contraventions and offences.*

Two exam points follow. First, the year is **2000**; later changes to the law were amendments to the 2000 Act, not a fresh Act, so an option calling the Act itself “the IT Act of 2008” is wrong. Second, the scope words are **electronic records, electronic governance, and cybercrime**: an option that limits the Act to hardware prices, typing tests, or office furniture has left the scope entirely. No section numbers are needed for this lesson; the exam fact is the Act's year and its three branches of coverage.

## 56.3 Malware: the umbrella and its members

**Malware**, short for malicious software, is any program written to harm a computer, steal data, or disrupt work. It is the umbrella term: the question stem may say “malware” when it means the general class, and name virus, worm, Trojan, or ransomware when it means one member. Two further nuisance programs belong nearby: **spyware**, which secretly watches the user's activity and reports it, and **adware**, which displays unwanted advertisements.

**Definition (Malware).** *Malicious software: any program designed to damage, disrupt, or gain unauthorised access to a computer system. Virus, worm, Trojan horse, and ransomware are types of malware.*

A **virus** attaches itself to a host file or program and spreads when that file is run or shared. It needs human action — opening an infected attachment, running an infected file, or carrying it on a pen drive — and once it runs it can corrupt, delete, or alter files. A **worm**, by contrast, is standalone: it copies itself and spreads across a network without any host file and without waiting for a user to click. Worms consume bandwidth and system resources, so their signature effect is a slowed or crashed network. A **Trojan horse** pretends to be useful software — a free utility, a game, a handy tool — but hides a harmful purpose inside. It does not replicate by itself; instead it opens a **backdoor** through which an attacker can steal passwords, spy on the user, or control the machine. **Ransomware** encrypts the victim's files or locks the screen and demands payment (a ransom) for release. Paying does not guarantee recovery; the reliable defence is a clean **backup** kept separate from the machine.

**Example 56.3.1.** *An item describes a program that “looks like a useful download, does not copy itself, but lets an attacker read passwords.” Trace the three contrasts: it needs no host file to be described (so not a virus), it does not self-spread (so not a worm), and it demands no payment (so not ransomware). The false-useful disguise plus the backdoor is the Trojan horse.*

## 56.4 Antivirus, quarantine, and patching

**Antivirus** software detects, blocks, and removes malware. It works by recognising known malware signatures and by flagging suspicious behaviour, and it must be updated regularly because new malware appears every day. When the antivirus meets a file it cannot yet judge, it places it in **quarantine**: an isolated holding area where the file cannot run or spread while the user or administrator decides its fate. Quarantine is isolation, not deletion and not a backup — the file is still present but harmless while isolated.

**Definition (Quarantine).** *The isolation of a suspicious file by security software so that it cannot execute or spread, pending inspection, removal, or restoration.*

**Patching** is the companion hygiene step: installing the vendor-supplied updates that fix known security flaws in the operating system and applications. An unpatched machine remains open to attacks that the vendor has already fixed, so “install updates promptly” is the exam’s standard correct advice. Antivirus cleans what is already inside, quarantine holds what is suspect, and patching closes the holes through which the next attack would enter.

## 56.5 Phishing and anti-phishing

**Phishing** is not malware at all but deception: a fake message — mail, SMS, or website — that impersonates a trusted sender (a bank, an office, a delivery firm) to steal passwords, OTPs, or money. Its warning signs are urgency and threat (“your account will be blocked”), unexpected attachments, links whose real destination does not match the claimed sender, and any request for a password or OTP. The defence, called **anti-phishing** practice, is procedural: verify the sender through a separate channel, hover over or long-press links to inspect them before clicking, never share an OTP or password by message, and report the message to the service provider or office.

**Definition (Phishing).** *A fraudulent message impersonating a trusted sender, designed to trick the recipient into revealing passwords, OTPs, or financial details.*

**Example 56.5.1.** *An item shows a mail reading “Dear customer, your account is blocked; click here and enter your OTP to restore it.” The correct response is to treat it as phishing: do not click, do not enter the OTP, and verify with the bank through its known number or site. A genuine bank or office never asks for a password or OTP by mail or message, so any option advising compliance is the trap.*

## 56.6 Encryption, digital signatures, and SSL/TLS

**Encryption** converts readable data (plaintext) into coded form (ciphertext) so that only authorised persons holding the key can read it. Its goal is confidentiality. A **digital signature**, by contrast, proves who sent a message and that it was not altered in transit; its goals are authenticity and integrity. The pair must never be swapped: a signature does not hide the message (anyone can still read it), and encryption by itself does not prove who sent it. This is the same distinction tested in the email lesson (TRAP-15), and it returns here because ransomware’s “encryption” step uses the same idea against the victim.

**SSL** (Secure Sockets Layer) and its successor **TLS** (Transport Layer Security) provide a secure, encrypted channel over a network. They protect data **in transit**: mail submission, web logins, and online forms, signalled by the padlock in the browser address bar. The lesson’s anchor fact, already seen in WO 2026 Q76, is that the secure email transmission protocol is **SSL/TLS**. Note the limit:

SSL/TLS protect the journey, not the endpoints — they do not clean an infected file and do not replace antivirus.

**Definition (Encryption).** *The conversion of plaintext into ciphertext with a key, so that only authorised holders of the key can recover the original content.*

## 56.7 Firewall, MFA, and the matching map

A **firewall** is a barrier, implemented in hardware or software (or both), that filters incoming and outgoing network traffic against a set of rules. It blocks unauthorised access while letting permitted traffic pass, like a gatekeeper at the office network door checking every packet against the list. Its partner inside the machine is the antivirus: the firewall filters traffic at the boundary, while the antivirus cleans malware already inside. One does not replace the other.

**MFA** (Multi-Factor Authentication) protects logins by demanding two or more independent checks — for example, a password plus an OTP, or a password plus a fingerprint. It closes the door against stolen passwords: even a leaked password is useless without the second factor. Strong, distinct passwords remain the first factor; MFA adds factors rather than replacing them.

| Control              | Job (the stem verb to watch)                          |
|----------------------|-------------------------------------------------------|
| Firewall             | Filters network traffic; blocks unauthorised access.  |
| Antivirus            | Detects and removes malware on the machine.           |
| Quarantine           | Isolates a suspicious file so it cannot run.          |
| Encryption / SSL-TLS | Hides content; protects data in transit.              |
| Digital signature    | Proves the sender and that content is intact.         |
| Patching             | Fixes known security flaws by installing updates.     |
| MFA                  | Adds extra login checks beyond the password.          |
| Backup               | Recovers data after ransomware, deletion, or failure. |

The matching technique is mechanical: read the verb in the stem. “Filters traffic” points to the firewall; “cleans malware” to the antivirus; “isolates” to quarantine; “hides content” to encryption; “proves sender” to digital signature; “fixes flaws” to patching; “extra login check” to MFA; “recovers after loss” to backup.

## 56.8 Exam retrieval and common confusions

Seven distinctions prevent most errors on this topic.

1. The IT Act is of **2000** and covers electronic records, e-governance, and cybercrime; later changes were amendments.
2. Malware is the **umbrella**; virus, worm, Trojan, and ransomware are specific types under it.
3. Virus needs a **host file plus human action**; worm is standalone and **self-spreading**; Trojan **does not replicate**; ransomware **locks files and demands payment**.
4. A digital signature **proves** (authenticity/integrity); it does not **hide** — encryption provides confidentiality (TRAP-15).
5. **SSL/TLS** protect data **in transit**; they do not clean infected files.

6. A **firewall** filters **traffic** at the boundary; antivirus cleans **malware** inside; quarantine **isolates** without deleting.
7. **MFA** adds login factors; **patching** fixes software flaws; **backup** is the recovery answer to ransomware.

**Example 56.8.1.** *An item asks which control “proves that a message came from the claimed sender and was not altered.” The verb “proves” selects the digital signature, not encryption (which hides) and not the firewall (which filters). If the stem instead says “hides the content from eavesdroppers,” the same map selects encryption; if it says “blocks unauthorised packets,” it selects the firewall.*

## Exercises

1. State the year and the three branches of scope of the Information Technology Act covered in this lesson.
2. Define malware and place virus, worm, Trojan horse, and ransomware under it in one sentence each.
3. Distinguish a virus from a worm on the two points of host file and human action.
4. Why is a Trojan horse dangerous even though it does not replicate?
5. Explain why backup, rather than payment, is the correct response to ransomware.
6. Contrast encryption with a digital signature: which goal does each serve?
7. Using the matching map, assign one control to each verb: filters, isolates, hides, proves, fixes.

## Solutions

1. The Information Technology Act is of **2000**. It gives legal validity to electronic records and electronic signatures, enables electronic governance, and defines cyber contraventions and offences.
2. Malware is malicious software written to harm or exploit. A virus attaches to a host file and needs human action; a worm self-copies across networks unaided; a Trojan disguises itself as useful software and opens a backdoor without replicating; ransomware locks or encrypts files and demands payment.
3. A virus needs a host file and a human act (opening, running, sharing) to spread; a worm is standalone, copies itself, and spreads over the network without a host file or user action.
4. The Trojan’s disguise persuades the user to install it, and the hidden backdoor then lets the attacker steal passwords, spy, or control the machine — damage that needs no replication.
5. Payment does not guarantee recovery and rewards the attacker, while a clean separate backup restores the data without negotiation; hence backup is the reliable defence.
6. Encryption serves **confidentiality** (hides content); a digital signature serves **authenticity and integrity** (proves sender and intactness).
7. Filters — firewall; isolates — quarantine; hides — encryption (SSL/TLS in transit); proves — digital signature; fixes — patching (installing updates).